BRAC University

Department of Mathematics and Natural Sciences

LECTURE ON

Real Analysis (MAT221)

Absolute and Conditional Convergence

Rearrangements, Double Summations and Products

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CONDUCTED BY

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Convergence of Infinite Series

Convergence of Infinite Series

We define the corresponding sequence of partial sums (s_m) by

$$s_m = a_1 + a_2 + a_3 + \cdots + a_m,$$

and say that the series $\sum_{n=1}^{\infty} a_n$ converges to S if the sequence (s_m) converges to S. In this case, we write

$$\sum_{n=1}^{\infty} a_n = S.$$

If the sequence $\{S_N\}$ does not converge, the series is called divergent.

Absolute Convergence

Absolute Convergence

A series $\sum_{n=1}^{\infty} a_n$ is said to be *absolutely convergent* if the series of absolute values $\sum_{n=1}^{\infty} |a_n|$ converges.

Theorem

Every absolutely convergent series is convergent.

Conditional Convergence

Conditional Convergence

A series $\sum_{n=1}^{\infty} a_n$ is said to be *conditionally convergent* if it converges but does not converge absolutely.

Problem

The alternating harmonic series

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots$$

is conditionally convergent.

Alternating Series Test

Let (a_n) be a sequence such that

1. $a_n \geq 0$ for all n,
2. $a_{n+1} \leq a_n$ for all n (i.e., (a_n) is non-increasing), and
3. $\lim_{n\to\infty} a_n = 0$.

Then the alternating series

$$\sum_{n=1}^{\infty} (-1)^{n+1} a_n$$

converges.

Problems with Conditional Convergence or Divergent Series

Rearrangements of Series

Rearrangement of a Series

A *rearrangement* of the series $\sum_{n=1}^{\infty} a_n$ is a series of the form

$$\sum_{n=1}^{\infty} a_{p(n)},$$

where $p : \mathbb{N} \to \mathbb{N}$ is a bijection.

Absolute Convergence and Rearrangements

Every rearrangement of an absolutely convergent series converges to the same sum as the original series.

Riemann Rearrangements Theorem

If a series $\sum_{n=1}^{\infty} a_n$ is conditionally convergent, then for any real number L, there exists a rearrangement of the series that converges to L. Furthermore, there exists a rearrangement that diverges.

Double Summations and Products

Double Series

A *double series* is a series of the form

$$\sum_{m=1}^{\infty} \sum_{n=1}^{\infty} a_{m,n},$$

where $a_{m,n}$ are real numbers indexed by two indices m and n.

Problem

Consider the doubly indexed family of real numbers $(a_{ij})_{i,j\in\mathbb{N}}$ defined by

$$a_{ij} = \begin{cases} -1, & \text{if } j = i, \\ \dfrac{1}{2^{j-i}}, & \text{if } j > i, \\ 0, & \text{if } j < i. \end{cases}$$

Equivalently, the infinite matrix (a_{ij}) has the form

$$\begin{pmatrix} -1 & \frac{1}{2} & \frac{1}{4} & \frac{1}{8} & \frac{1}{16} & \cdots \\ 0 & -1 & \frac{1}{2} & \frac{1}{4} & \frac{1}{8} & \cdots \\ 0 & 0 & -1 & \frac{1}{2} & \frac{1}{4} & \cdots \\ 0 & 0 & 0 & -1 & \frac{1}{2} & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots & \ddots \end{pmatrix}.$$

We wish to attach a mathematical meaning to the formal double series

$$\sum_{i=1}^{\infty}\sum_{j=1}^{\infty} a_{ij}.$$

Fubini's Theorem for Series

If the double series $\sum_{m=1}^{\infty} \sum_{n=1}^{\infty} |a_{m,n}|$ converges, then the iterated sums

$$\sum_{m=1}^{\infty} \left(\sum_{n=1}^{\infty} a_{m,n} \right) \quad \text{and} \quad \sum_{n=1}^{\infty} \left(\sum_{m=1}^{\infty} a_{m,n} \right)$$

converge and are equal to the same value as the double series.

Product of Series

Given two series $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$, their *Cauchy product* is defined as the series

$$\sum_{n=1}^{\infty} c_n,$$

where

$$c_n = \sum_{k=1}^{n} a_k b_{n-k+1}.$$

Mertens' Theorem

If $\sum_{n=1}^{\infty} a_n$ converges to A and $\sum_{n=1}^{\infty} b_n$ converges to B, and at least one of these series converges absolutely, then their Cauchy product converges to AB.

Thank You!

We'd love your questions and feedback.

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References

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[2] Terence Tao, *Analysis I*, 3rd Edition, Texts and Readings in Mathematics, Hindustan Book Agency, 2016.